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Age-Critical Joint Communication and Computation Offloading for Satellite-Integrated Internet

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***Abstract*—The emerging sixth-generation (6G) networks are expected to deliver ubiquitous intelligent services through satellite-integrated Internet, addressing the stringent demands of computation-intensive and delay-sensitive applications in extreme and infrastructure-sparse environments. Existing offloading approaches often fail to achieve age-critical resource allocation in dynamic satellite-integrated Internet, potentially leading to outdated information and inefficient resource utilization. To address this, we propose an age-critical joint communication and computation offloading scheme, where low Earth orbit satellites cover several base stations and many ground users. Then, we formulate a comprehensive optimization problem, which aims to minimize the average peak age of information (PAoI), subject to energy consumption constraints. We propose an online Lyapunov-enhanced decoupled actor-critic (LDA) scheme. Specifically, our LDA scheme decomposes the original long-term problem into manageable per-frame subproblems. It employs a deep neural network integrated with an order-preserving quantization approach to effectively decouple offloading decisions, and utilizes low-complexity iterative algorithms for resource allocation. Extensive simulations demonstrate that our LDA scheme substantially reduces the average PAoI compared to the related deep reinforcement learning schemes, while satisfying energy consumption constraints and reducing up to 25% complexity.**

***Index Terms*—Satellite-integrated Internet, Low Earth orbit satellite, Lyapunov optimization, resource allocation.**

I. Introduction

Satellite-integrated Internet has emerged as a critical enabler for delivering ubiquitous intelligent services in the sixth-generation (6G) networks, particularly addressing the stringent requirements of computation-intensive and delay-sensitive applications in extreme and infrastructure-sparse environments [1], [2]. Among various satellite technologies, low Earth orbit satellites (LEOS) have attracted particular attention due to their advantages such as extensive coverage and low propagation latency, enhancing connectivity and computational capabilities of satellite-integrated Internet [3], [4]. Moreover, the unprecedented satellite density yields rich spatial diversity [5], e.g., a ground user equipment (UE) could simultaneously observe about 150 satellites of the Starlink constellation phase I, which provide the potential of multi-satellite coordinated services. Within the satellite-integrated Internet, joint communication and computational offloading significantly enhances the quality of service of UEs [6], by effectively distributing tasks and leveraging the complementary computational resources of both terrestrial base stations (BSs) and the LEOSs [7]. This capability thereby greatly enhances the ability to support emerging applications, remote health care and smart transportation, along with diverse needs across multiple sectors [8]. However, emerging applications such as disaster assessment, smart traffic management, and emergency response generate vast amounts of sensing data that frequently surpass the computational capabilities of terrestrial BSs alone, necessitating support from LEOS for handling such demanding tasks [9], [10].

Critically, existing offloading approaches often fail to achieve age-critical resource allocation of emerging applications in satellite-integrated Internet [11], [12]. Traditional metrics like average delay inadequately reflect the worst case latency and cannot ensure timely processing of information updates, potentially leading to outdated information and inefficient resource utilization [13]. Furthermore, conventional methods including those relying on fixed positions of UEs, or heuristic resource allocation struggle to manage the inherent mobility and variability of satellite-integrated Internet [14]. Moreover, approaches based on deep reinforcement learning (DRL) typically require extensive offline training and struggle with online adaptation [15], [16]. These limitations are exacerbated where dynamic topologies and strict onboard constraints demand real time responsiveness and robustness. To address these gaps, we propose an online joint communication and computation offloading scheme for satellite-integrated Internet in this paper, explicitly designed to achieve age-critical while satisfying energy constraints.

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A. Related Works

To optimize the computational and communication performance of UEs in satellite-integrated Internet, extensive research has been conducted on opportunistic computation offloading approaches. These approaches typically involve solving mixed-integer nonlinear programming (MINLP) problems to jointly determine binary offloading decisions, and allocate communication and computational resources [7]. Solving such problems often requires substantial computational complexity, consequently, various suboptimal algorithms with reduced complexity have been developed [17]. However, these methods face significant practical challenges, such as inconsistent convergence performance in heuristic algorithms and growing approximation gaps under dynamic network conditions [18]. Particularly in satellite-integrated Internet, traditional approaches that rely on fixed satellite positions cannot effectively manage the inherent mobility and variability of LEOS [19]. Furthermore, existing distributed offloading frameworks often assume abundant and consistent satellite computational resources, which leads to degraded performance under realistic and dynamic operating conditions [20], [21].

Most offloading schemes optimize average delay, which does not capture information freshness for time sensitive applications [22], [23]. The age of information (AoI) quantifies information freshness as the elapsed time from generation to successful processing and provides a rigorous timeliness measure for scheduling and resource allocation [24], [25]. Further, AoI not only measures the timeliness of tasks but also incorporates scheduling strategies that account for queuing backlogs, encouraging the system to prioritize remaining tasks [26], which effectively prevents the accumulation of task data that could not be completed within the prescribed time frame. In satellite integrated Internet, for time sensitive sensing and control scenarios, adopting AoI as the performance metric better addresses coverage gaps and backlog accumulation caused by intermittent satellite visibility, satellite handovers, long propagation with rain attenuation, and multi frame queueing and computation [27]. By capturing freshness across frames rather than only per packet delay, AoI guides offloading and resource allocation toward timely updates, thereby improving the delivery of computation intensive and low latency services.

Furthermore, guaranteeing system stability with respect to both data queue lengths and energy consumption constitutes a critical objective within this online joint communication and computation offloading scheme [28]. Lyapunov optimization provides a robust methodological framework for addressing this challenge [29], which facilitates the decomposition of long-term stochastic optimization problems into deterministic single-term subproblems, thereby delivering rigorous theoretical guarantees for system stability [30], [31]. Nevertheless, when DRL algorithms are applied within the Lyapunov framework, achieving effective online learning performance proves difficult [32]. A primary cause for this difficulty is the inherent reliance of DRL on exploration during its initial learning phase, which frequently results in suboptimal resource allocation decisions [33], [34]. The inefficiency associated with this initial phase induces rapid accumulation of task data within queues over sustained operation, and such instability represents a fundamental limitation for deployment within highly dynamic satellite-integrated Internet [35]. Moreover, a principal drawback is their characteristically high computational complexity, stemming from the requirement to solve challenging MINLP subproblems at each decision epoch [36]. This requirement inevitably creates an intractable tradeoff between computational tractability and system performance.

B. Contributions

Motivated by these critical gaps, we propose a long-term, age-critical joint communication and computation offloading framework in this paper. Specifically, we leverage Lyapunov optimization to decompose complex long-term optimization problems into manageable per-frame subproblems, ensuring long-term system stability. Then, we introduce an online Lyapunov-enhanced decoupled actor-critic (LDA) scheme, which efficiently decouples offloading decisions from computational resource allocation through deep neural networks (DNN) and order-preserving quantization, thereby significantly reducing computational complexity while maintaining age-critical performance. The main contributions of this paper are summarized as follows:

- To address the inherent stochasticity introduced by satellite-integrated Internet mobility and random task arrivals, and further consider the time-coupled nature of task processing across consecutive time frames, we propose an age-critical joint communication and computation offloading framework. This framework introduces a weighted PAoI metric and a cross-frame task-processing mechanism, which facilitates the low-complexity pursuit of globally optimal resource allocation of long-term task offloading optimization, while rigorously capturing the tasks timeliness.
- We employ the Lyapunov optimization framework to solve the formulated long-term stochastic optimization problem, which is intractable and requires a robust mechanism for online decision-making. Specifically, by introducing dedicated virtual queues for task data backlog and energy consumption, we decompose the original long-term task offloading optimization problem into a sequence of solvable deterministic subproblems. Furthermore, detailed theoretical derivations and analysis are provided to ensure the feasibility and effectiveness of our decoupling approach.
- We propose the Lyapunov-enhanced Decoupled Actor-Critic (LDA) scheme for the subproblem in each frame, which can solve the coupled optimization of discrete offloading decisions and continuous resource allocations. The LDA introduces a two-stage solution, where a Deep Neural Network (DNN) integrated with an order-preserving quantization mechanism efficiently determines the high-dimensional discrete offloading policy. Subsequently, by fixing the offloading decisions, the remaining continuous resource allocation problem is significantly simplified and solved using low-complexity iterative algorithms. Simulation results demonstrate that our LDA

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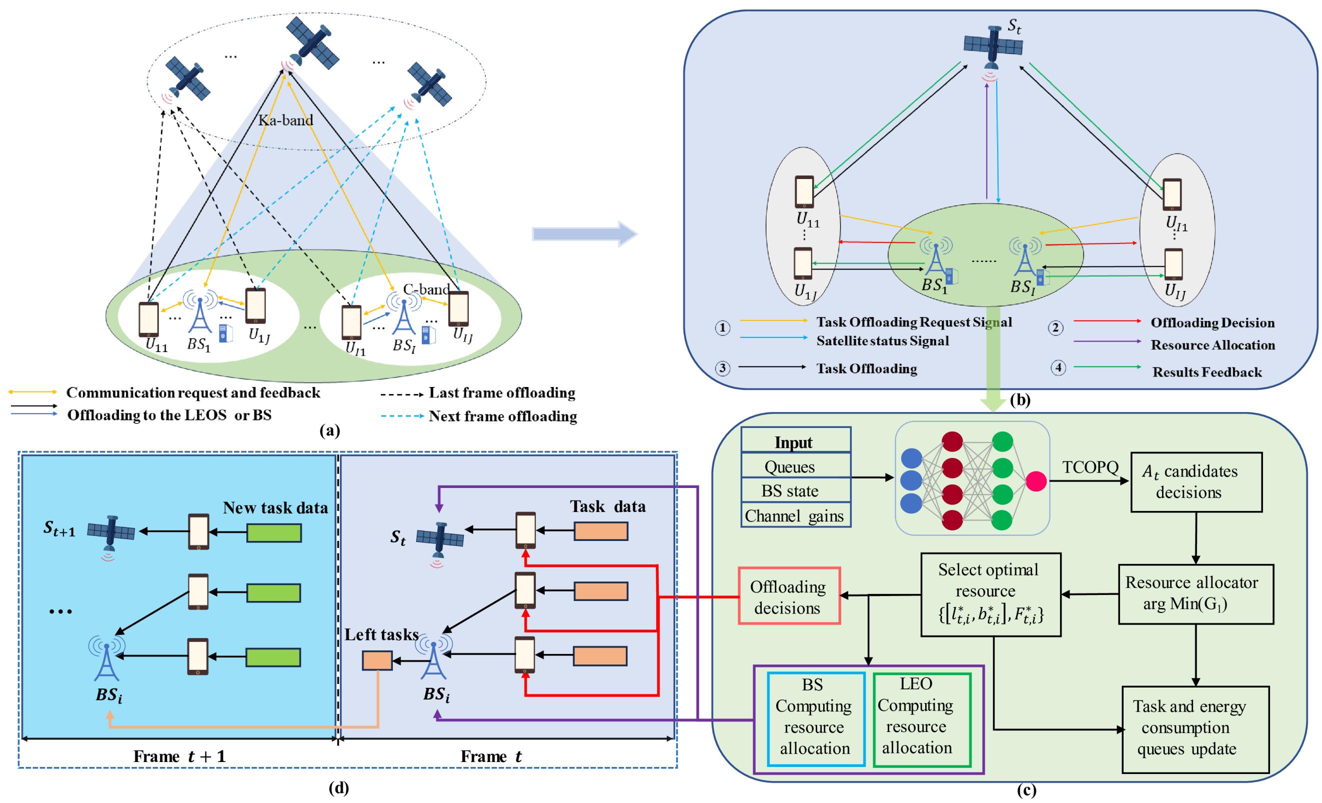

Fig. 1. Age-critical joint communication and computation offloading scheme to optimize the average PAoI in satellite-integrated Internet: (a) Long-term comprehensive task offloading scenario for LEOSs assisted in satellite-integrated Internet; (b) Offloading access scheme including the request, decision, task processing and results feedback; (c) Offloading framework for each BS jointly generates offloading decisions, and computing resource allocation; (d) Multiple-frame task processing with residual tasks carried forward to subsequent frames for task completion.

scheme achieves minimal average PAoI, faster convergence and reduces complexity by up to 25% compared to the related AC algorithms [37] and BS supported schemes [12].

The rest of the paper is organized as follows. Section II presents the system model, featuring a long-term task offloading framework. Section III details the Lyapunov decomposition that transforms the long-term problem into single-frame subproblems. Section IV introduces the online LDA scheme, which decouples task offloading from resource allocation through noisy order-preserving quantization. Section V evaluates our LDA scheme against related offloading schemes, emphasizing improvements in convergence and optimization effectiveness. Finally, Section VI concludes the paper and outlines future research directions.

II. System Model

In this section, we present an age-critical joint communication and computation offloading scheme for satellite-integrated Internet.

A. Task Offloading Optimization Framework

Fig. 1 illustrates our long-term age-critical joint communication and computation offloading scheme in satellite-integrated Internet. Each UE employs a single antenna, whereas each BS and each LEOS are with multi-antennas and onboard computational servers for task execution. Orthogonal frequency-division multiple access (OFDMA) is adopted to avoid intra-band interference. As shown in Fig. 1(a), each UE can choose between two offloading modes: offloading to the BS over the C band (4-8 GHz), which provides moderate bandwidth and low rain attenuation and is suitable for terrestrial access, or offloading to the LEOS over the Ka band (26.5-40 GHz), which offers wider contiguous spectrum and supports higher data rates at the cost of stronger rain attenuation and greater free space path loss [38], [39]. Once the tasks are processed at the LEOS or BS, the computing results are fed back to the UEs via the respective links.

In our scheme, we consider a LEOS in conjunction with I BSs in each time frame, where each BS serves J terrestrial UEs, and the LEOS can effectively support $I \cdot J$ UEs. The sets of BSs and UEs in each BS are denoted by $\mathcal{I} = \{1, \ldots, i, \ldots, I\}$ and $\mathcal{J} = \{1, \ldots, j, \ldots, J\}$, and the ith BS and its jth UE are denoted as BS_i and U_{ij}, respectively, where $i \in \mathcal{I}$, $j \in \mathcal{J}$. Let $\mathcal{T} = \{1, \ldots, T\}$ denote the set of time frames, where all frames have a common duration τ. Our scheme assumes that at the beginning of each frame $t \in \mathcal{T}$, $I \cdot J$ UEs generate a sequence of tasks, defined as $\mathcal{L} = \{\mathbf{L}_1, \ldots, \mathbf{L}_t, \ldots, \mathbf{L}_T\}$, and in frame t, the task numbers are represented by $\mathbf{L}_t = [L_{t,ij}]_{I \times J}$, where $L_{t,ij}$ denotes data size in bits of the task generated by U_{ij} in frame t. To flexibly model the processing, we permit a tasks processing time to

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span multiple frames, allowing a residual portion to carry over from frame t to frame $t+1$. The total processing time for any single task is strictly bounded within $(0, 2\tau)$, which is based on the assumption that system parameters are configured for a steady state operation [12], [31], and any task needing more than two full frames is treated as an extreme overload [40].

This multi-frame processing capability necessitates careful satellite handover management. At the beginning of each frame $t \in \mathcal{T}$, a handover may occur. Based on the ephemeris information, the BSs initialize the candidate satellite set $\mathcal{V}_t$. To support the potential multi-frame task duration, $\mathcal{V}_t$ includes only LEO satellites that are visible during frame t, have no unfinished task carried over from prior frames, and have a predicted visibility span of at least two frames. The BSs then select the serving satellite S_t from $\mathcal{V}_t$, and broadcast this selection, upon which UEs associate with S_t for uplink offloading. Newly generated offloaded tasks are sent to S_t, whereas tasks left unfinished in frame $t-1$ continue to be processed on S_{t-1} until completion.

Assume tasks are indivisible [41], [42], each task must be executed at exactly one of three locations: UE, BS, or LEOS. We use $l_{t,ij} \in \{0,1\}$ to denote whether the task is processed locally or offloaded, and $b_{t,ij} \in \{0,1\}$ (valid when $l_{t,ij} = 0$) to specify the offloading destination. Thus, the joint decision can be expressed as

$$(l_{t,ij},\, b_{t,ij}) = \begin{cases} (1,0), & \text{task } L_{t,ij} \text{ is processed at } U_{ij}, \\ (0,1), & \text{task } L_{t,ij} \text{ is offloaded to } BS_i, \\ (0,0), & \text{task } L_{t,ij} \text{ is offloaded to } S_t. \end{cases} \tag{1}$$

B. Communication Model

The UEs can offload their task either to the BSs or to the LEOS for processing when the computational capability of the terrestrial UEs can not meet their task processing requirements. To meet the computation target of terrestrial UEs, we propose a joint offloading scheme in a single time frame as shown in Fig.1(b), where the implementation of the offloading scheme involves four sequential operations: Information submission, decision making, task offloading and processing, and result feedback.

1) Offload to BS: We define a set of UEs to BS distances over long-term frames as $\mathcal{D} = \{\mathbf{D}_1, \ldots, \mathbf{D}_t, \ldots, \mathbf{D}_T\}$, where $\mathbf{D}_t = [d_{t,ij}^{BS_i}]_{I \times J}$ and $d_{t,ij}^{BS_i}$ denotes the distance between U_{ij} and BS_i in frame t, and the distances follow a random distribution across different time frames. Assuming line-of-sight (LoS) propagation between the UEs and BSs in frame t, and the channel gain $h_{t,ij}^{BS_i}$ can be expressed as [43]:

$$h_{t,ij}^{BS_i} = \frac{\lambda_C}{4\pi d_{t,ij}^{BS_i}}, \tag{2}$$

where λ_C is the carrier wavelength of the C-band, and $h_{t,ij}^{BS_i}$ affects the offloading decision. From this, the transmission rate from U_{ij} to BS_i in frame t as:

$$R_{t,ij}^{BS_i} = \frac{B_C}{J}\log_2\left(1 + \frac{p_{ij}\left|h_{t,ij}^{BS_i}\right|^2}{\sigma_1^2}\right), \tag{3}$$

where B_C is the total bandwidth in the C-band, p_{ij} is the transmit power of U_{ij}, σ_1^2 is the noise power, which is an additive white Gaussian noise (AWGN) at BS_i. From this, the transmission delay of U_{ij} offloading tasks to BS_i in frame t as:

$$T_{t,ij}^{BS_i,tran} = \frac{L_{t,ij} b_{t,ij}\left(1 - l_{t,ij}\right)}{R_{t,ij}^{BS_i}}, \forall t,i,j \in \mathcal{T},\mathcal{I},\mathcal{J}. \tag{4}$$

2) Offload to LEOS: For the UEs to LEOS communications, consider that the altitude, speed and position of the LEOS are known to all BSs in each frame due to the ephemeris [44]. We assume that the distance between UEs and BS is negligible, and thus the communication distance between all UEs and the selected LEOS can be approximated as equal in each frame. In our long-term age-critical joint communication and computation offloading scheme, the BSs select a LEOS without offloading tasks for the covered USs to handover in each frame. Therefore, the distance between UEs and the selected LEOS is changing over different frames. Therefore, we denote the UE to LEOS distance sequence across frames by $D^S = \{d_1, \ldots, d_t, \ldots, d_T\}$, where d_t is the distance between UEs and the selected LEOS S_t in frame t.

Assume that the UE to LEOS links are utilized the widely recognized shadowed-Rician fading channel [45], which can effectively account for both fading and masking influences. Additionally, each UE to LEOS link follows independent and identically distributed (i.i.d.), with the probability density function (PDF) of shadowed-Rician fading channel gain as followed [45]:

$$\begin{aligned} f_{|h_{t,ij}|^2}(x) = &\left(\frac{2b_{t,ij}m_{t,ij}}{2b_t m_{t,ij} + \Omega_{t,ij}}\right)^{m_{t,ij}} \frac{1}{2b_{t,ij}} \exp\left(-\frac{x}{2b_{t,ij}}\right) \\ &\cdot {}_1F_1\left(m_{t,ij}, 1, \frac{\Omega_{t,ij}x}{2b_{t,ij}\left(2b_{t,ij}m_{t,ij} + \Omega_{t,ij}\right)}\right), \end{aligned} \tag{5}$$

where $2b_{t,ij}$ represents the mean power of multi path components, ${}_1F_1(\cdot,\cdot,\cdot)$ denotes the confluent hypergeometric function, $m_{t,ij}$ indicates the Nakagami-m parameter, and $\Omega_{t,ij}$ stands for the mean power of the LoS component. Moreover, the small-scale fading coefficient $|h_{t,ij}|$ is obtained using the accept-reject sampling method [46]. The path loss combines $|h_k|$ with the LoS propagation, and the complete channel gain is given by

$$h_{t,ij}^{S_t} = |h_k| \cdot \frac{\lambda_{Ka}}{4\pi (d_t)^\beta}, \tag{6}$$

where λ_{Ka} is the carrier wavelength of the Ka-band and β is the path loss exponent. From this, the transmission rate of U_{ij} offloading tasks to the LEOS as:

$$R_{t,ij}^{S_t} = \frac{B_{Ka}}{IJ}\log_2\left(1 + \frac{p_{ij}\left|h_{t,ij}^{S_t}\right|^2}{\sigma_2^2}\right), \tag{7}$$

where B_{Ka} is the total bandwidth in the Ka-band and σ_2^2 is the noise power of AWGN at LEOS. The transmission delay of U_{ij} offloading tasks to S_t as:

$$T_{t,ij}^{S_t,tran} = \frac{L_{t,ij}\left(1 - b_{t,ij}\right)\left(1 - l_{t,ij}\right)}{R_{t,ij}^{S_t}}, \forall t,i,j \in \mathcal{T},\mathcal{I},\mathcal{J}. \tag{8}$$

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C. Computing Model

Departing from the single frame processing requirement in prior works [47], [48], we allow the process of tasks to span into multiple frames, yielding a more flexible offloading model. As shown in Fig. 1(c), each BS employs an offloading framework that jointly determines the execution location of tasks and the allocation of computing resources for its associated users. Consider that each task is executed at UEs, BS, or LEOS, and the analysis examines three resulting cases.

1) Local Computing: We denote the local CPU frequency of U_{ij} as f_{ij}^{l}, which is measured in CPU cycles per second. Let ϕ represent the number of CPU cycles required to process one bit of data [49], because the task is processed locally, the time for carrying out the local computing in frame t is

$$T_{t,ij}^{l,proc} = \frac{\phi l_{t,ij} L_{t,ij}}{f_{ij}^{l}}, \forall t,i,j \in \mathcal{T},\mathcal{I},\mathcal{J}. \tag{9}$$

Each UE executes the following local rule: if $T_{t,ij}^{l,proc} > \tau$, we have $l_{t,ij} = 0$ and the task will be offloaded to the BS or LEOS with stronger computing capabilities; otherwise, we have $l_{t,ij} = 1$ and the task is locally computed at the UE.

2) BS Computing: As shown in Fig. 1(d), tasks that are not completed within a frame are carried forward to subsequent frames and continue to be processed at the node where they were started. Unfinished tasks from the previous frame increase the delay in the current frame, so we analyze the effects at the BS as follows.

At frame t, BS_i may process two queues: Resident left tasks $L_{t-1,ij}^{BS_i,left}$ and new arrivals $L_{t,ij}$, which first experience a transmission delay $T_{t,ij}^{BS_i,tran}$. To process the remaining tasks, BS_i allocates computational resources $f_{t,ij}^{\prime BS_i}$ ($f_{1,ij}^{\prime BS_i} = 0$) (in CPU cycles per second) to process the remaining tasks $L_{t-1,ij}^{BS_i,left}$. Let $f_{max}^{BS_i}$ denote the maximum computational capability at the BS_i, and $\sum_{j=1}^{J_{t-1}^{BS_i}} f_{t,ij}^{\prime BS_i} \leq f_{\max}^{BS_i}$ needs to be satisfied [12]. The required time to process these tasks is:

$$T_{t-1,ij}^{BS_i,left} = \frac{\phi L_{t-1,ij}^{BS_i,left}}{f_{t,ij}^{\prime BS_i}}, \tag{10}$$

and we define the maximum processing time for the remaining tasks across all UEs as:

$$T_{t-1}^{BS_i,left} = \max_{j \in J}\{T_{t-1,ij}^{BS_i,left}\}. \tag{11}$$

After completing the remaining tasks, BS_i begins processing the newly offloaded tasks. Since the new tasks must wait for completing transmission, the actual start time for processing depends on the transmission delay $T_{t,ij}^{BS_i,tran}$ and the remaining task processing time $T_{t-1}^{BS_i,left}$. Therefore, the available time for U_{ij} to process new tasks in frame t is:

$$T_{t,ij}^{BS_i,proc} = \left[\tau - \max\{T_{t,ij}^{BS_i,tran}, T_{t-1}^{BS_i,left}\}\right]^{+}, \tag{12}$$

where $[x]^{+} \triangleq \max\{0, x\}$.

Within the available time $T_{t,ij}^{BS_i,proc}$, BS_i processes the new task $L_{t,ij}$ using computational capacity $f_{t,ij}^{BS_i}$. Also the condition $\sum_{j=1}^{J} f_{t,ij}^{BS_i} \leq f_{\max}^{BS_i}$ needs to be satisfied. The amount of data processed is:

$$L_{t,ij}^{BS_i,proc} = \min\{L_{t,ij} b_{t,ij} (1 - l_{t,ij}), \frac{f_{t,ij}^{BS_i} T_{t,ij}^{BS_i,proc}}{\phi}\}, \forall t,i,j \in \mathcal{T},\mathcal{I},\mathcal{J}. \tag{13}$$

If the task is not fully processed at BS_i, the remaining task data volume is:

$$L_{t,ij}^{BS_i,left} = L_{t,ij} b_{t,ij} (1 - l_{t,ij}) - L_{t,ij}^{BS_i,proc}. \tag{14}$$

Unfinished task $L_{t,ij}^{BS_i,left}$ move to frame $t+1$, where BS_i assigns $f_{t+1,ij}^{\prime BS_i}$ and spends $T_{t,ij}^{BS_i,\text{left}}$ to $L_{t,ij}^{BS_i,left}$. Fig. 2 depicts this recurring workflow, which is repeated each frame for new and residual tasks.

It should be noted that the processing delay at the BS includes computation and the uplink communication, while the feedback delay is negligible because the computational results are relatively small [12].

3) LEOS Computing: Compared to offloading to BS, since the transmission distance between the UE and LEOS is significantly increased, when UEs offload tasks to LEOS, it is necessary to additionally consider the propagation delay, and we have:

$$T_{t,ij}^{prop} = \frac{c\,(1 - b_{t,ij})\,(1 - l_{t,ij})}{d_t}, \forall t,i,j \in \mathcal{T},\mathcal{I},\mathcal{J}, \tag{15}$$

where c is speed of light.

We assume that a new LEOS without any computing task is selected at the beginning of frame t, and any unfinished task in frame $t-1$ remains on the last LEOS until completion. Thus, the available time for $L_{t,ij}$ in frame t is:

$$T_{t,ij}^{S_t,proc} = [\tau - T_{t,ij}^{S_t,tran} - T_{t,ij}^{prop}]^{+}. \tag{16}$$

When $L_{t,ij}$ is offloaded to S_t, let $f_{\max}^{S_t}$ and $f_{t,ij}^{S_t}$ denote the maximum computational capability at the LEOS and the computational capability of every LEOS allocated to U_{ij}, the

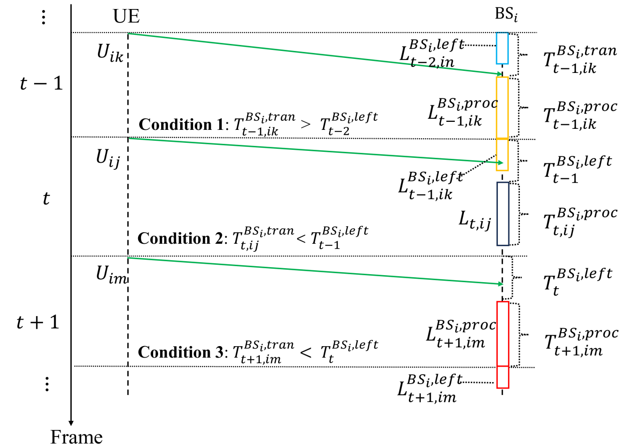

Fig. 2. Task offloading and processing between UEs and BS_i across multiple frames.

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condition $\sum_{j=1}^{J} f_{ij}^{S_t} \le f_{\max}^{S_t}$ needs to be satisfied. The amount of processed data is:

$$L_{t,ij}^{S_t,proc} = \min\{L_{t,ij}(1-b_{t,ij})(1-l_{t,ij}), \frac{f_{t,ij}^{S_t} T_{t,ij}^{S_t,proc}}{\phi}\}, \forall t,i,j \in \mathcal{T},\mathcal{I},\mathcal{J}. \tag{17}$$

If the task is not fully processed at S_t, the remaining task data volume is:

$$L_{t,ij}^{S_t,left} = L_{t,ij}(1-b_{t,ij})(1-l_{t,ij}) - L_{t,ij}^{S_t,proc}. \tag{18}$$

At the end of frame t, the remaining tasks $L_{t,ij}^{S_t,left}$ are carried over to frame $t+1$. At the beginning of frame $t+1$, the LEOS used in frame t continues to allocate resources $f_{t+1,ij}^{\prime S_t}$ to process these tasks, the required time is

$$T_{t,ij}^{S_t,left} = \frac{\phi L_{t,ij}^{S_t,left}}{f_{t+1,ij}^{\prime S_t}}. \tag{19}$$

Also, the feedback delay is negligible because the computational results are relatively small.

D. Energy Model

1) BS Energy: As stated in [50], the energy consumption of a CPU cycle is given by κf^2, where κ is a constant related to the hardware architecture. Hence, the total energy consumed $E_t^{BS_i}$ at BS_i during the frame t can be expressed as:

$$E_t^{BS_i} = \sum_{j=1}^{J} \kappa_1 \phi((f_{t,ij}^{\prime BS_i})^2 L_{t-1,ij}^{BS_i,left} + (f_{t,ij}^{BS_i})^2 L_{t,ij}^{BS_i,proc}), \tag{20}$$

where κ_1 denotes the computing energy efficiency of BSs.

2) LEOS Energy: Due to the LEOS handover occurring at the beginning of each frame, tasks that could not be completed within the frame $t-1$ will continue to be executed on S_{t-1}. The energy dissipation required for processing at S_t in frame t a are given by:

$$E_t^{S_t} = \sum_{i=1}^{I}\sum_{j=1}^{J} \kappa_2 \phi (f_{t,ij}^{S_t})^2 L_{t,ij}^{S_t,proc}, \tag{21}$$

where κ_2 denotes the computing energy efficiency of LEOSs, and for processing the residual tasks on S_{t-1} carried over from frame $t-1$ is given by:

$$E_t^{S_{t-1}} = \sum_{i=1}^{I}\sum_{j=1}^{J} \kappa_2 \phi (f_{t,ij}^{\prime S_{t-1}})^2 L_{t-1,ij}^{S_{t-1},left}. \tag{22}$$

Main notations are summarized in Table I.

TABLE I
LIST OF NOTATIONS

Symbol	**Meaning**
I	Number of base stations (BSs)
J	Number of UEs per BS
U_{ij}	UE j associated with BS_i
S_t	Visible LEOS in frame t
t, τ	Frame index; frame duration
λ_C	Carrier wavelengths of C-band link
λ_{Ka}	Carrier wavelengths of Ka-band link
B_C	Total available bandwidth over C-band
B_{Ka}	Total available bandwidth over Ka-band
$h_{t,ij}^{BS_i}$, $h_{t,ij}^{S_t}$	Channel gains
$l_{t,ij} \in \{0,1\}$	Local execution decision
$b_{t,ij} \in \{0,1\}$	Offloading destination decision
$(L_{t,ij})_{\text{Location}}$	Location indicator vector
$L_{t,ij}$	Task data (in bits) generated by U_{ij} in frame t
$R_{t,ij}^{BS_i}$, $R_{t,ij}^{S_t}$	Achievable rates on U_{ij} to BS_i and S_t links
$T_{t,ij}^{BS_i,tran}$, $T_{t,ij}^{S_t,tran}$	Transmission delays of U_{ij} to BS_i and S_t
$L_{t,ij}^{BS_i,proc}$, $L_{t,ij}^{S_t,proc}$	Processed data size at BS_i and at S_t in frame t
$f_{t,ij}^{l}$, $f_{t,ij}^{BS_i}$, $f_{t,ij}^{S_t}$	Allocated CPU frequency at UE, BS_i, and S_t
$T_{t,ij}^{BS_i,proc}$, $T_{t,ij}^{S_t,proc}$	Available time to process tasks at BS_i and S_t
$T_{t,ij}^{BS_i,left}$, $T_{t,ij}^{S_t,left}$	Delays of left tasks at BS_i and S_t
ϕ	The number of CPU cycles per bit
$E_t^{BS_i}$, $E_t^{S_t}$	The energy consumed at BS_i and S_t

III. Problem Formulation

A. Problem Formulation

For long-term age-critical joint communication and computation offloading, we introduce a weighted AoI to enforce stricter latency control. The AoI is defined in a task-centric manner, where each task maintains an independent AoI. Let θ denote the continuous time instant, for the new tasks in frame t, the weighted AoI defined by

$$\text{AoI}_t(\theta) = \big(\theta - (t-1)\tau\big) + w\big[\theta - t\tau\big]^+, \tag{23}$$

where $[x]^+ \triangleq \max\{0,x\}$ and $w > 1$ is a penalty factor. Because the downlink transmission delay is negligible, while the millisecond-level propagation delay is insignificant compared with the frame duration. Thus, the AoI of a task resets to zero upon completion. This weighting drives the scheduler to minimize peak AoI $P_{t,ij}$ for U_{ij} in frame t, we have:

$$P_{t,ij} = T_{t,ij}^{l,proc} + (\tau - T_{t,ij}^{S_t,proc} + \frac{\phi L_{t,ij}^{S_t,proc}}{f_{t,ij}^{S_t}} + wT_{t,ij}^{S_t,left}) + (\tau - T_{t,ij}^{BS_i,proc} + \frac{\phi L_{t,ij}^{BS_i,proc}}{f_{t,ij}^{BS_i}} + wT_{t,ij}^{BS_i,left}). \tag{24}$$

Note that (24) includes all three execution modes, the offloading decisions have been embedded in the definitions of the involved terms. Therefore, only one group of terms is active for a given task, ensuring that each task is executed at exactly one location.

We aim for minimizing the long-term average PAoI, while constraining the energy consumption of BS and LEOS by jointly optimizing the offloading decisions $\mathbf{b} = \{\mathbf{b}_t\}_{t=1}^{T}$ and $\mathbf{l} = \{\mathbf{l}_t\}_{t=1}^{T}$, the BS computing resource allocation $\mathbf{F}^{BS} = \{\mathbf{F}_t^{BS}\}_{t=1}^{T}$ and $\mathbf{F}^{\prime BS} = \{\mathbf{F}_t^{\prime BS}\}_{t=1}^{T}$, the LEOS computing resource allocation $\mathbf{F}^{LEOS} = \{\mathbf{F}_t^{S}\}_{t=1}^{T}$ and $\mathbf{F}^{\prime S} = \{\mathbf{F}_t^{\prime S}\}_{t=1}^{T}$. Let $\mathbf{b}_t = [b_{t,ij}]_{I\times J}$, $\mathbf{F}_t^{BS} = [f_{t,ij}^{BS}]_{I\times J}$, $\mathbf{F}_t^{\prime BS} = [f_{t,ij}^{\prime BS}]_{I\times J}$, $\mathbf{F}_t^{S} = [f_{t,ij}^{S_t}]_{I\times J}$, $\mathbf{F}_t^{\prime S} = [f_{t,ij}^{\prime S_t}]_{I\times J}$, respectively. Let $\mathbf{F} =$

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$\{\mathbf{F}^{BS}, \mathbf{F}'^{BS}, \mathbf{F}^{S}, \mathbf{F}'^{S}\}$. Therefore, a long-term average PAoI minimization problem is formulated as follows:

$$\mathcal{P}0: \min_{\mathbf{b},\mathbf{l},\mathbf{F}} \lim_{T\to\infty} \frac{1}{T}\sum_{t=1}^{T}\frac{1}{IJ}\sum_{i=1}^{I}\sum_{j=1}^{J} P_{t,ij} \tag{25}$$

$$\text{s.t.} \quad l_{t,ij} \in \{0,1\}, \forall t \in \mathcal{T}, \forall i \in \mathcal{I}, \forall j \in \mathcal{J}, \tag{25a}$$

$$b_{t,ij} \in \{0,1\}, \forall t \in \mathcal{T}, \forall i \in \mathcal{I}, \forall j \in \mathcal{J}, \tag{25b}$$

$$\sum_{j=1}^{J}(1-l_{t,ij})b_{t,ij}f_{t,ij}^{BS_i} \le f_{\max}^{BS_i}, \tag{25c}$$

$$\sum_{j=1}^{J}(1-l_{t,ij})b_{t,ij}f_{t+1,ij}'^{BS_i} \le f_{\max}^{BS_i}, \tag{25d}$$

$$\sum_{i=1}^{I}\sum_{j=1}^{J}(1-l_{t,ij})(1-b_{t,ij})f_{t,ij}^{S_t} \le f_{\max}^{S_t}, \tag{25e}$$

$$\sum_{i=1}^{I}\sum_{j=1}^{J}(1-l_{t,ij})(1-b_{t,ij})f_{t+1,ij}'^{S_t} \le f_{\max}^{S_t}, \tag{25f}$$

$$\lim_{T\to\infty}\frac{1}{T}\cdot\sum_{t=1}^{T}E_t^{BS_i} \le E_{\max}^{BS_i}, \forall i \in \mathcal{I}, \tag{25g}$$

$$E_t^{S_t} \le E_{\max}^{S_t},\ E_t^{S_{t-1}} \le E_{\max}^{S_{t-1}}, \forall t \in \mathcal{T}, \tag{25h}$$

$$f_{t,ij}^{BS_i}, f_{t,ij}'^{BS_i}, f_{t,ij}^{S_t}, f_{t,ij}'^{S_t} \ge 0. \tag{25i}$$

The constraints in Problem $\mathcal{P}0$ are detailed as follows: (25a) and (25b) indicate the offloading decisions, (25c) and (25d) indicate that the total computational capability assigned to BS_i must not exceed its limit $f_{\max}^{BS_i}$, (25e) and (25f) indicate that the total computational capability assigned to S_t must not exceed $f_{\max}^{S_t}$, (25g) indicate that the long-term average energy consumed by BS_i has to be less than $E_{max}^{BS_i}$, (25h) indicate that the energy consumption consumed by S_t and the energy consumption for processing the residual tasks on S_{t-1} has to be less than $E_{\max}^{S_t}$ and $E_{\max}^{S_{t-1}}$.

B. Lyapunov-based Decoupling

We apply the Lyapunov optimization to decouple $\mathcal{P}0$ into subproblems in each frame. The transformation of long-term constraints into manageable entities is achieved through the establishment of virtual queues $\mathbf{Q}(t)$, $\mathbf{E}(t)$. Specifically, task queues are instituted for U_{ij}, with their evolution governed by:

$$Q_{ij}(t+1) = Q_{ij}(t) + L_{t,ij} - L_{t-1,ij}^{left} - L_{t,ij}^{proc}, \tag{26}$$

where $Q_{ij}(1) = 0$, $L_{0,ij}^{left} = 0$. Concurrently, energy queues are formulated for BS_i articulated as:

$$E_i(t+1) = E_i(t) + E_t^{BS} - E_{\max}^{BS}, \tag{27}$$

where $E_i(1) = 0$. Intuitively, when the virtual energy queues are stable, the average energy does not exceed $E_{\max}^{BS_i}$, and thus the constraints in (25g) are satisfied.

By introducing the combined backlog vector $\mathbf{X}(t) = \{\mathbf{Q}(t), \mathbf{E}(t)\}$ for joint dynamics of data and energy queues, where $\mathbf{Q}(t)$ and $\mathbf{E}(t)$ denote $\{Q_{ij}(t)\}_{i=1,j=1}^{I,J}$ and $\{E_i(t)\}_{i=1}^{I}$, respectively. Following the Lyapunov optimization framework [29], we define $V(\mathbf{X}(t))$ as the Lyapunov function and $\Delta V(\mathbf{X}(t))$ as its drift:

$$\begin{aligned} V(\mathbf{X}(t)) &= 0.5\left(\sum_{i=1}^{I}\sum_{j=1}^{J}Q_{ij}(t)^2 + \sum_{i=1}^{I}E_i(t)^2\right), \\ \Delta V(\mathbf{X}(t)) &= \mathbb{E}\{V(\mathbf{X}(t+1)) - V(\mathbf{X}(t)) \mid \mathbf{X}(t)\}. \end{aligned} \tag{28}$$

To minimize the PAoI while stabilizing $\mathbf{X}$, we define the drift-plus-penalty expression (DPE) as:

$$\Gamma(\mathbf{X}(t)) \triangleq \Delta V(\mathbf{X}(t)) + K\cdot\sum_{i=1}^{I}\sum_{j=1}^{J}\mathbb{E}\left\{P_{t,ij} \mid \mathbf{X}(t)\right\}, \tag{29}$$

where K is a weight to scale the penalty. Specifically, we seek to minimize an upper bound of the DPE. First, we derive an upper bound of $\Gamma(\mathbf{X}(t))$ and let:

$$V(\mathbf{Q}(t)) \triangleq 0.5\sum_{i=1}^{I}\sum_{j=1}^{J}Q_{ij}(t)^2, \tag{30}$$

$$\Delta V(\mathbf{Q}(t)) \triangleq \mathbb{E}\{V(\mathbf{Q}(t+1)) - V(\mathbf{Q}(t)) \mid \mathbf{X}(t)\}. \tag{31}$$

We can get:

$$\Delta V(\mathbf{Q}(t)) \le C_1 + \sum_{i=1}^{I}\sum_{j=1}^{J}Q_{ij}(t)\left(L_{t,ij} - L_{t-1,ij}^{left} - L_{t,ij}^{proc}\right), \tag{32}$$

where the constant C_1 is obtained as:

$$C_1 = 0.5\sum_{i=1}^{I}\sum_{j=1}^{J}(\max\{L_{t,ij}, L_{t-1,ij}\})^2. \tag{33}$$

The detail derivations of (32) and (33) are presented in Appendix A.

Similarly, we define:

$$V(\mathbf{E}(t)) \triangleq 0.5\sum_{i=1}^{I}E_i(t)^2, \tag{34}$$

$$\Delta V(\mathbf{E}(t)) \triangleq \mathbb{E}\{V(\mathbf{E}(t+1)) - V(\mathbf{E}(t)) \mid \mathbf{X}(t)\}, \tag{35}$$

we can get:

$$\Delta V(\mathbf{E}(t)) \le C_2 + \sum_{i=1}^{I}E_i(t)\left(E_t^{BS_i} - E_{\max}^{BS_i}\right), \tag{36}$$

where the constant C_2 is obtained as:

$$C_2 = 0.5\sum_{i=1}^{I}(\max\{(\kappa_1(f_{\max}^{BS_i})^3\tau)^2, (E_{\max}^{BS_i})^2\}). \tag{37}$$

The derivations of (36) and (37) follow the similar procedure to that of Appendix A, and thus is omitted here for conciseness.

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Summing over the two inequalities in (32) and (36), we have the upper bound of DPE in (29) as

$$C + \sum_{i=1}^{I}\sum_{j=1}^{J} Q_{ij}(t)\mathbb{E}\left[\left(L_{t,ij} - L_{t-1,ij}^{left} - L_{t,ij}^{proc}\right) \mid \mathbf{X}(t)\right] + \sum_{i=1}^{I} E_i(t)\mathbb{E}\left[\left(E_t^{BS_i} - E_{\max}^{BS_i}\right) \mid \mathbf{X}(t)\right] + K\sum_{i=1}^{I}\sum_{j=1}^{J}\mathbb{E}\left\{P_{t,ij} \mid \mathbf{X}(t)\right\}, \tag{38}$$

where $C = C_1 + C_2$. By eliminating constant terms, we simplify (38) as follows:

$$G_1(t) = \sum_{i=1}^{I}\sum_{j=1}^{J}\left(Q_{ij}(t)\left(L_{t-1,ij}^{proc} - L_{t,ij}^{proc}\right) + KP_{t,ij}\right) + \sum_{i=1}^{I} E_i(t)E_t^{BS_i}. \tag{39}$$

By utilizing the Lyapunov optimization framework to decouple $\mathcal{P}0$, it can be transformed into the solutions of a series of single-frame optimization problems $\mathcal{P}1$ as follows:

$$\mathcal{P}1: \min_{\mathbf{b}_t, \mathbf{l}_t, \mathbf{F}_t} \quad G_1(t) \tag{40}$$

$$\text{s.t.} \quad l_{t,ij} \in \{0,1\}, \forall i \in \mathcal{I}, \forall j \in \mathcal{J}, \tag{40a}$$

$$b_{t,ij} \in \{0,1\}, \forall i \in \mathcal{I}, \forall j \in \mathcal{J}, \tag{40b}$$

$$(25c) - (25f), (25h), (25i) \tag{40c}$$

Problem $\mathcal{P}1$ is still challenging to solve because it is an MINLP problem, in the following section, we propose an online DRL-based algorithm to solve $\mathcal{P}1$ efficiently.

IV. Decoupled Offloading and Resource Optimization

Fig. 3 outlines a two-layer process of our LDA scheme: A DNN outputs an initial offloading vector, which is quantized to yield several candidates. Then, we solve a resource-allocation subproblem for each candidate, evaluate the global index $G_1(t)$, and keep the candidate with the lowest $G_1(t)$. Thus, our LDA jointly selects the offloading decisions and its optimal computing resources in one pass.

A. Description of LDA Scheme

At each frame t, BS_i inputs the state $X_{t,i} = \{Q_{ij}(t), E_i(t), T_{t-1}^{BS_i,left}, R_{t,ij}^{BS_i}, R_{t,ij}^{S_t}\}_{j=1}^{J}$ to a DNN network with parameters $\theta_{t,i}$ (randomly initialized when $t = 1$). The sigmoid layer produces $\tilde{\mathbf{b}}_{t,i} \in [0,1]^J$, where $\tilde{b}_{t,ij}$ is the offload probability for U_{ij}. Applying the thresholding operator $\mathcal{Q}(\cdot)$ binarizes $\tilde{\mathbf{b}}_{t,i}$ into the decision vector $\mathbf{b}_{t,i}$, we have

$$b_{t,ij}^1 = \mathcal{Q}(\tilde{b}_{t,ij}) = \begin{cases} 1, & \text{if } \tilde{b}_{t,ij} \geq 0.5, \\ 0, & \text{if } \tilde{b}_{t,ij} < 0.5. \end{cases} \tag{41}$$

At each frame t, the DNN outputs an offloading vector $\mathbf{b}_{t,i}$ that is binarized by $\mathcal{Q}(\cdot)$ to yield the action $\tilde{\mathbf{b}}_{t,i}$. Environment feedbacks from this action updates $\theta_{t,i}$. To hasten convergence, we also generate extra candidate vectors and sample them to balance exploration and exploitation.

Here, we apply the threshold-constrained order-preserving quantization (TCOPQ) method to generate A_t candidate decisions. In noisy order-preserving quantization (OPQ) [51], the elements of the continuous relaxation vector $\tilde{\mathbf{b}}_{t,i}$ are sorted based on their absolute differences from the threshold value 0.5. Moreover, we set a dynamic threshold δ_t to adaptively adjust the generation of candidate decisions A_t, such that:

$$|\tilde{b}_{t,i(1)} - 0.5| \leq \cdots |\tilde{b}_{t,i(j)} - 0.5| \leq \cdots \leq |\tilde{b}_{t,i(k)} - 0.5| \leq \delta_t, \tag{42}$$

where $((1), \ldots, (j), \ldots, (k))$ representing the permuted indices after sorting and $A_t = k$.

Then, $\tilde{b}_{t,i(a)}$ are the decisions thresholds to quantize $\tilde{\mathbf{b}}_{t,i}$, where the a-th action $\mathbf{b}_{t,i}^a$ for $a = 2, ..., A_t$, is obtained from entry-wise comparisons of and that is:

$$b_{t,ij}^a = \begin{cases} 1, & \tilde{b}_{t,ij} > \tilde{b}_{t,i(j-1)} \\ & \text{or } (\tilde{b}_{t,ij} = \tilde{b}_{t,i(j-1)} \wedge \tilde{b}_{t,i(j-1)} \leq 0.5), \\ 0, & \tilde{b}_{t,ij} < \tilde{b}_{t,i(j-1)} \\ & \text{or } (\tilde{b}_{t,ij} = \tilde{b}_{t,i(j-1)} \wedge \tilde{b}_{t,i(j-1)} > 0.5). \end{cases} \tag{43}$$

Each candidate $\mathbf{b}_{t,i}^a$ is ranked by analytically solving its resource-allocation subproblem to minimize $G_1(t)$, and the best one $\mathbf{b}_{t,i}^*$ is executed, giving precise feedback that speeds convergence.

Evaluating every one of A_t candidates boosts accuracy but quickly becomes expensive. We curb this cost with an adaptive margin δ_t: Only vectors satisfying $|\tilde{b}_{t,ij} - 0.5| \leq \delta_t$ enter the resource-allocation step. The margin starts at $\delta_1 = 0.5$ and decays multiplicatively every ΔT frames, dynamically balancing solution quality and runtime.

$$\delta_t = \begin{cases} \max(\gamma\delta_{t-1}, 0.1), & \text{if } \mathrm{mod}(t, \Delta T_A) = 0, \\ \delta_{t-1}, & \text{if } \mathrm{mod}(t, \Delta T_A) \neq 0, \end{cases} \tag{44}$$

where $\gamma \in (0,1)$ is a scaling factor.

Our LDA scheme keeps the latest M state action pairs $(X_{t,i}, \mathbf{b}_{t,i}^*)$ in a replay buffer $\mathcal{D}$. Once $|\mathcal{D}| \geq M/2$, the DNN is retrained every ΔT_L frames, at frames $t \equiv 0 \pmod{\Delta T_L}$

$$G_2^i(t) = \sum_{j=1}^{J}\left(-Q_{ij}(t)L_{t,ij}^{BS_i,proc} + K\left(\frac{\phi L_{t,ij}^{BS_i,proc}}{f_{t,ij}^{BS_i}} + wT_{t,ij}^{BS_i,left}\right) + E_i(t)\kappa_1\phi\left(f_{t,ij}^{BS_i}\right)^2 L_{t,ij}^{BS_i,proc}\right). \tag{47}$$

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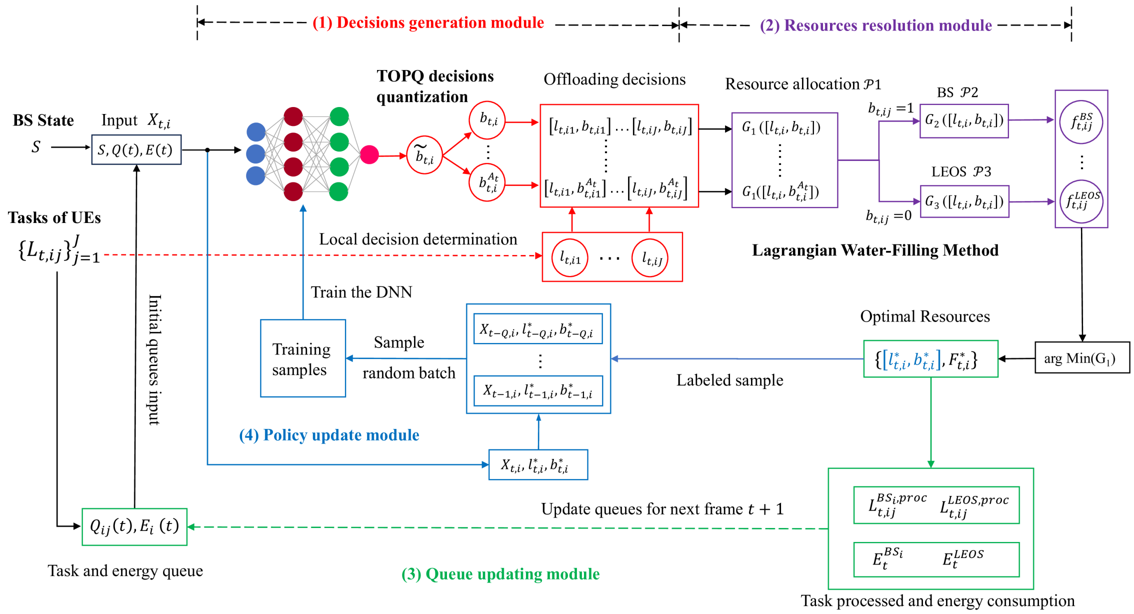

Fig. 3. The schematics of the proposed LDA scheme

we sample a minibatch $\mathcal{C} \subset \mathcal{D}$ and update $\theta_{t,i}$ by minimizing a focal cross-entropy loss.

$$\mathcal{L}_{\mathcal{S}}(\boldsymbol{\theta}_{t,i}) = -\frac{1}{|\mathcal{C}|}\sum_{\in\mathcal{C}_i}\sum_{j=1}^{J}\Big(b^*_{t,ij}\log(\tilde{b}_{t,ij}) \\ +(1-b^*_{t,ij})\log(1-\tilde{b}_{t,ij})\Big). \tag{45}$$

LDA outputs the joint action $(\mathbf{l}_t^*, \mathbf{b}_t^*)$ and resource map $\mathbf{F}_t^*$, which are executed to process tasks; BS and LEOS energy costs follow (20) and (21). At frame $t+1$ the system updates: Task queues Q_{ij}, energy queues E_i, and BS carry-over time $T^{BS_i,left}$ via (26), (27), and (11). New channel rates $\{R^{BS_i}_{t+1,ij}, R^{S_{t+1}}_{t+1,ij}\}$ are observed, forming the next state vector $X_{t+1,i} = \{Q_{ij}(t+1), E_i(t+1), T_t^{BS_i,left}, R^{BS_i}_{t+1,ij}, R^{S_{t+1}}_{t+1,ij}\}_{j=1}^{J}$. This feeds the DNN for the next iteration.

We summarize the LDA scheme in Algorithm 1, where the major computational complexity is in line 9 that computes $G_1(t)$ by solving the optimal resource allocation problems. In the next subsection, we propose the low complexity algorithms to obtain optimal computing resource allocation.

B. Optimal Resource Allocation Algorithms

Given the value of $\{\mathbf{l}_t, \mathbf{b}_t\}$ in $\mathcal{P}1$, it can be reformulated as:

$$\mathcal{P}2: \min_{\mathbf{F}_t}\ G_1(t) \tag{46}$$

$$\text{s.t.}\quad (25c)-(25f), (25h), (25i) \tag{46a}$$

We can observe that Problem $\mathcal{P}2$ is a decoupling problem, since the BS offloading and the LEOS offloading are decoupled with the given offloading volume matrix $\mathbf{L}$ and the offloading decisions $\{\mathbf{l}_t, \mathbf{b}_t\}$. Problem $\mathcal{P}2$ can be decomposed into the following two sub-problems: The computing resource optimization of each BS and the LEOS, respectively.

1) Optimization of the BS Computing Resource: Decoupling $\mathcal{P}2$ yields independent BS and LEOS tasks. For BS_i at frame t, treat $L^{\text{proc}}_{t-1,ij}$ and $Q_{ij}(t)$ as constants, gather $\mathbf{F}_t$ dependent terms, and reformulate the objective as $G_2^i(t)$ in Eq. (47) at the bottom of this page. This gives the BS subproblem $\mathcal{P}3$ as follows,

$$\mathcal{P}3: \min_{\mathbf{F}_t^{BS_i}}\ G_2^i(t) \tag{48}$$

$$\text{s.t.}\quad (25c), (25d), \tag{48a}$$

$$f^{BS_i}_{t,ij} \geq 0, f'^{BS_i}_{t+1,ij} \geq 0. \tag{48b}$$

To solve $\mathcal{P}3$, we classify UEs as two types. One is called type $\mathcal{A}$ if its entire task can be finished within $T^{BS_i,proc}_{t,ij}$,

$$\frac{\phi L_{t,ij}}{f^{BS_i}_{t,ij}} \leq T^{BS_i,proc}_{t,ij} \iff f^{BS_i}_{t,ij} \geq f^{\text{th}}_{t,ij} \triangleq \frac{\phi L_{t,ij}}{T^{BS_i,proc}_{t,ij}}, \tag{49}$$

and type $\mathcal{B}$ otherwise. Define the index sets $\mathcal{A}_{t,i}(f) = \{j \mid f^{BS_i}_{t,ij} \geq f^{\text{th}}_{t,ij}\}$, $\mathcal{B}_{t,i}(f) = \mathcal{J} \setminus \mathcal{A}_{t,i}(f)$.

For the optimal computing resources allocation in processing $L^{BS_i,left}_{t,ij}$ for $\mathcal{B}_{t,i}(f)$, a proportional allocation strategy can be derived based on the Cauchy-Schwarz inequality. In the next frame, the whole CPU budget $f^{BS_i}_{\max}$ of BS_i is proportionally shared by the UEs in $\mathcal{B}_{t,i}(f)$,

$$f'^{BS_i}_{t+1,ij} = f^{BS_i}_{\max}\frac{L^{BS_i,left}_{t,ij}}{L^{left}_{t,i}(f)}, L^{left}_{t,i}(f) \triangleq \sum_{k\in\mathcal{B}} L^{BS_i,left}_{t,ik}. \tag{50}$$

This proportional allocation strategy yields an aggregate processing time mathematically equivalent to the total serial

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Algorithm 1: The LDA Scheme for Offloading and Resource Optimization

Initialization:
- Initialize DNN parameters $\boldsymbol{\theta}_{1,i}$ with random weights
- Initialize empty experience buffer $\mathcal{D}$
- Initialize task queues $Q_{ij}(1) = 0, \forall i, j$
- Initialize energy queues $E_i(1) = 0, \forall i$
- Initial time budgets $T_0^{BS_i,left} = 0, \forall i$
- Define training batch size $|\mathcal{B}_i|, \forall i$
- Initial $\delta_1 = 0.5$, γ, ΔT_A, ΔT_L

for *each frame* $t = 1, 2, \ldots, T$ **do**
  Observe system state $X_{t,i} = \{Q_{ij}(t), E_i(t), T_{t-1}^{BS_i,left}, R_{t,ij}^{BS_i}, R_{t,ij}^{S_t}\}_{i=1}^{J}$
  Generate offloading decisions $\mathbf{l}_{t,i}^{*}$ by Eq.(9)
  Generate relaxed offloading decisions $\tilde{\mathbf{b}}_{t,i} = \mathrm{DNN}_{\boldsymbol{\theta}_{t,i}}(X_{t,i})$ using sigmoid output
  Apply TCOPQ quantization:
    Sort $\tilde{b}_{t,ij}$ by $|\tilde{b}_{t,ij} - 0.5|$ ascending
    Generate A_t candidates $\{\mathbf{b}_{t,i}^{a}\}$ via threshold δ_t using Eq.(43)
  for *each candidate* $a = 1 \to A_t$ **do**
    Solve computing resource allocation $\mathbf{F}_{t,i}^{a*}$ using **Algorithm 2** and **Algorithm 3** with $\mathbf{b}_{t,i}^{a}$
    Calculate performance metric $G_1^{a}(t)$ using Eq.(39)
  end
  Select optimal action: $\mathbf{b}_{t,i}^{*} = \arg\min_a G_1^{a}(t)$
  Execute $\mathbf{b}_{t,i}^{*}$ and $\mathbf{F}_{t,i}^{*}$, update experience buffer $\mathcal{D} \leftarrow \mathcal{D} \cup (X_{t,i}, \mathbf{b}_{t,i}^{*})$
  if *mod*$(t, \Delta T_A) = 0$ **then**
    Update threshold: $\delta_t = \max(\gamma\delta_{t-1}, 0.1)$
  end
  if $|\mathcal{D}| \geq Q/2 \wedge$ *mod*$(t, \Delta T_L) = 0$ **then**
    Sample mini-batch $\mathcal{C}_i$ from $\mathcal{D}$
    Update DNN parameters via Adam optimizer: $\boldsymbol{\theta}_{t+1,i} \leftarrow \boldsymbol{\theta}_{t,i} - \eta\nabla\mathcal{L}_{\mathcal{S}}(\boldsymbol{\theta}_{t,i})$
  end
  Update system states for $t+1$:
  - Update task queues $Q_{ij}(t+1)$ via Eq.(26)
  - Update energy queues $E_i(t+1)$ via Eq.(27)
  - Update time budgets $T_t^{BS_i,left}$ via Eq.(11)

end

Output: Optimal joint decisions and computing resource allocations $\{\mathbf{l}_{t,i}^{*}, \mathbf{b}_{t,i}^{*}, \mathbf{F}_{t,i}^{*}\}$ for each frame

processing delay. Therefore, the total residual processing time is

$$\sum_{k\in\mathcal{B}} T_{t,ik}^{BS_i,left} = \frac{\phi}{f_{\max}^{BS_i}}\Big[\sum_{k\in\mathcal{B}} L_{t,ik} - \sum_{k\in\mathcal{B}}\frac{f_{t,ik}^{BS_i} T_{t,ik}^{BS_i,proc}}{\phi}\Big]. \quad (51)$$

Putting the above pieces into $G_2^i(t,f)$ yields

$$\begin{aligned} G_2^i(t,f) = &\sum_{n\in\mathcal{A}}\Big[-Q_{in}(t)L_{t,in} + E_i(t)\kappa_1\phi L_{t,in}(f_{t,in}^{BS_i})^2 \\ &+K\frac{\phi L_{t,in}}{f_{t,in}^{BS_i}}\Big] + \sum_{k\in\mathcal{B}} T_{t,ik}^{BS_i,proc}\Big[-\Big(\frac{Q_{ik}(t)}{\phi} + M\Big)f_{t,ik}^{BS_i} \\ &+E_i(t)\kappa_1(f_{t,ik}^{BS_i})^3\Big] + |\mathcal{B}_{t,i}(f)|\,K\tau + M\sum_{k\in\mathcal{B}} L_{t,ik}, \end{aligned} \quad (52)$$

where $M = \frac{Kw}{f_{\max}^{BS_i}}$. Owing to the piecewise-convex structure of (52), the optimisation remains convex. We adopt *water-filling* [52] method to solve this problem, and formulate the Lagrangian functions as

$$\mathcal{L}_{t,i}(f,\lambda) = G_2^i(t,f) + \lambda\sum_{j=1}^{J}(f_{t,ij}^{BS_i} - f_{\max}^{BS_i}). \quad (53)$$

Algorithm 2: The computing resource optimization of BS for $\mathcal{P}3$

Input: $\{L_{t,ij}, T_{t,ij}^{BS_i,proc}, Q_{t,ij}(t)\}_{j=1}^{J}$, $E_i(t)$, κ_1, K, w, ϕ, τ, $f_{\max}^{BS_i}$

$\lambda_{\mathrm{L}} \leftarrow 0,\ \lambda_{\mathrm{H}} \leftarrow \lambda_{\max}$;
while $|\lambda_H - \lambda_L| > \varepsilon$ **do**
  $\lambda \leftarrow (\lambda_{\mathrm{H}} + \lambda_{\mathrm{L}})/2$;
  foreach $j \in \mathcal{J}$ **do**
    compute $f_{t,ij}^{A}(\lambda)$, by *Newton*;
    compute $f_{t,ij}^{B}(\lambda)$, by (56);
    select $f_{t,ij}^{BS_i}(\lambda)$ with piecewise rule (57);
  end
  $R_{t,i}(\lambda) \leftarrow \sum_j f_{t,ij}^{BS_i}(\lambda)$;
  if $R_{t,i}(\lambda) > f_{\max}^{BS_i}$ **then** $\lambda_{\mathrm{L}} \leftarrow \lambda$;
  else $\lambda_{\mathrm{H}} \leftarrow \lambda$;
end
$\{f_{t,ij}^{BS_i\star}\} \leftarrow \{f_{t,ij}^{BS_i}(\lambda_{\mathrm{high}})\}$;
classify $\mathcal{A}^\star, \mathcal{B}^\star$ by $f_{t,ij}^{BS_i\star}$ and $f_{t,ij}^{\mathrm{th}}$;

Output: $\mathcal{A}^\star, \mathcal{B}^\star, \{f_{t,ij}^{BS_i}\}$

By setting the derivative of Lagrangian component associated with constraint $\mathcal{A}$ to zero, we obtain:

$$h_{t,in}^{A}(f) = 2E_i(t)\kappa_1\phi L_{t,in}(f_{t,in}^{BS_i})^3 + \lambda(f_{t,in}^{BS_i})^2 - K\phi L_{t,in} = 0. \quad (54)$$

This constitutes a strictly increasing cubic equation that guarantees a unique positive root. Starting with the initial approximation $f_{t,in}^{(0)} = \left(\frac{K}{2E_i(t)\kappa_1\phi L_{t,in}}\right)^{1/3}$, we implement *Newton-Raphson* iterations as follows,

$$f_{t,in}^{(q+1)} = f_{t,in}^{(q)} - \frac{h_{t,in}^{A}(f_{t,in}^{(q)})}{6E\kappa_1\phi L_{t,in}(f_{t,in}^{(q)})^2 + 2\lambda f_{t,in}^{(q)}}. \quad (55)$$

Numerical experiments demonstrate that three to five iterations typically suffice to achieve a highly accurate solution. We denote the result value as $f_{t,in}^{A}(\lambda)$.

The derivative of $\mathcal{B}$ part yields a quadratic $3E_i(t)\kappa_1 T_{t,ik}^{BS_i,proc}(f_{t,ik}^{BS_i})^2 - \big(\frac{Q_{ik}(t)}{\phi} + M\big)T_{t,ik}^{BS_i,proc} + \lambda = 0$, whose non-negative root is

$$f_{t,ik}^{B}(\lambda) = \sqrt{\frac{\big(Q_{ik}(t)/\phi + M\big)}{3E_i(t)\kappa_1} - \frac{\lambda}{3E_i(t)\kappa_1 T_{t,ik}^{BS_i,proc}}}. \quad (56)$$

For given λ pick

$$f_{t,ij}^{BS_i}(\lambda) = \begin{cases} f_{t,ij}^{B}(\lambda), & f_{t,ij}^{B}(\lambda) < f_{t,ij}^{\mathrm{th}}, \\ f_{t,ij}^{A}(\lambda), & f_{t,ij}^{B}(\lambda) \geq f_{t,ij}^{\mathrm{th}}. \end{cases} \quad (57)$$

Define the monotone load $R_{t,i}(\lambda) = \sum_j f_{t,ij}^{BS_i}(\lambda)$. Bisection on $\lambda \in [0, \lambda_{\max}]$ ($\lambda_{\max}$ is restricted by (56) finds the unique $\lambda^\star$ satisfying $R_{t,i}(\lambda^\star) = f_{\max}^{BS_i}$.

Based on the above discussions, Algorithm 2 presents the solution of Problem $\mathcal{P}3$.

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2) Optimization of the LEOS Computing Resource: We reformulate the LEOS optimization objective at frame t by isolating terms explicitly dependent on $\mathbf{F}_t^S$ as:

$$\mathcal{P}4 : \min_{\mathbf{F}_t^S} \ G_3(t) \tag{58}$$

$$\text{s.t.} \quad (25e), (25f), (25h) \tag{58a}$$

$$f_{t,ij}^{S_t} \geq 0, \ f_{t+1,ij}^{\prime S_t} \geq 0. \tag{58b}$$

The simplified objective $G_3(t)$ is explicitly expressed as:

$$G_3(t) = \sum_{i=1}^{I} \sum_{j=1}^{J} \Big(-Q_{ij}(t) L_{t,ij}^{S_t,proc} + K\big(\frac{\phi L_{t,ij}^{S_t,proc}}{f_{t,ij}^{S_t}} + w T_{t,ij}^{S_t,left}\big)\Big). \tag{59}$$

The definitions for UE types ($\mathcal{A}'$, $\mathcal{B}'$) and the next frame resource allocation approaches follow those previously outlined for BS resources, with the following adjustments:

- Thresholds and residual computations use LEOS-specific parameters $f_{\max}^{S_t}$ and $T_{t,ij}^{S_t,proc}$.
- Energy-related constraints differ, involving new parameters such as κ_2, and potentially distinct Lagrangian multipliers (μ, ν).

Moreover, the energy consumption at LEOS in frame $t+1$ incurred by unfinished tasks must be explicitly accounted for $E_{\max}^{S_t}$. The energy constraints of unfinished tasks (25h) can be satisfied by adopting a resource allocation mechanism analogous to (50).

Thus, $G_3(t,f)$ yields

$$\begin{aligned} G_3(t,f) = & \sum_{i=1}^{I} \sum_{n' \in \mathcal{A}'} \Big[-Q_{in'}(t) L_{t,in'} + K \frac{\phi L_{t,in'}}{f_{t,in'}^{S_t}} \Big] \\ & - \sum_{i=1}^{I} \sum_{k' \in \mathcal{B}'} T_{t,ik'}^{S_t,proc} \Big(\frac{Q_{ik'}(t)}{\phi} + M' \Big) f_{t,ik'}^{S_t} \\ & + \sum_{i=1}^{I} \left| \mathcal{B}'_{t,i}(f) \right| K\tau + M' \sum_{i=1}^{I} \sum_{k' \in \mathcal{B}'} L_{t,ik'}, \end{aligned} \tag{60}$$

where $M' = \frac{Kw}{f_{\max}^{S_t}}$. Consequently, the Lagrangian function is:

$$\begin{aligned} \mathcal{L}_t(f,\mu,\nu) = & G_3(t,f) + \mu \sum_{i=1}^{I} \sum_{j=1}^{J} f_{t,ij}^{S_t} - f_{\max}^{S_t} \\ & + \nu \sum_{i=1}^{I} \Bigg(\sum_{n' \in \mathcal{A}'} \kappa_2 \phi (f_{t,in'}^{S_t})^2 L_{t,in'} \\ & + \sum_{k' \in \mathcal{B}'} \kappa_2 (f_{t,ik'}^{S_t})^3 T_{t,ik'}^{S_t,proc} \Bigg). \end{aligned} \tag{61}$$

By setting the derivative of the Lagrangian component associated with constraint $\mathcal{A}'$ to zero, we obtain:

$$h_{t,in'}^{A'}(f) = 2\kappa_2 \phi \nu L_{t,in'} (f_{t,in'}^{S_t})^3 + \mu (f_{t,in'}^{S_t})^2 - K\phi L_{t,in'}. \tag{62}$$

This constitutes a strictly increasing cubic equation that guarantees a unique positive root. Starting with the initial approximation $f_{t,in'}^{(0)} = \left(\frac{K}{2\kappa_2 \phi \nu L_{t,in'}} \right)^{1/3}$, we implement *Newton-Raphson* iterations:

$$f_{t,in'}^{(q'+1)} = f_{t,in'}^{(q')} - \frac{h_{t,in'}^{A'}(f_{t,in'}^{(q')})}{6\kappa_2 \phi \nu L_{t,in'} (f_{t,in'}^{(q')})^2 + 2\mu f_{t,in'}^{(q')}}, \tag{63}$$

Numerical experiments demonstrate that three to five iterations typically suffice to achieve a highly accurate solution. We denote the result value as $f_{t,in'}^{A'}(\mu,\nu)$.

The derivative of $\mathcal{B}'$ part yields a quadratic $3\kappa_2 \nu T_{t,ik'}^{S_t,proc} (f_{t,ik'})^2 - (\frac{Q_{ik'}(t)}{\phi} + M') T_{t,ik'}^{S_t,proc} + \mu = 0$ whose non-negative root is

$$f_{t,ik'}^{B'}(\mu,\nu) = \sqrt{\frac{(Q_{ik'}(t)/\phi + M')}{3\kappa_2 \nu} - \frac{\mu}{3\kappa_2 \nu T_{t,ik'}^{S_t,proc}}}. \tag{64}$$

For given μ, ν pick

$$f_{t,ij}^{LEOS}(\mu,\nu) = \begin{cases} f_{t,ij}^{B'}(\mu,\nu), & f_{t,ij}^{B'}(\mu,\nu) < f_{t,ij}^{\prime \text{th}}, \\ f_{t,ij}^{A'}(\mu,\nu), & f_{t,ij}^{B'}(\mu,\nu) \geq f_{t,ij}^{\prime \text{th}}. \end{cases} \tag{65}$$

Define the monotone load

$$R_t'(\mu,\nu) = \sum_{i=1}^{I} \sum_{j} f_{t,ij}^{S_t}(\mu,\nu). \tag{66}$$

Bisection on $\mu \in [0, \mu_{\max}]$ and $\nu \in [0, \nu_{\max}]$, $\mu_{\max}$ is restricted by (64), $\nu_{\max}$ is restricted by (62) ($\nu_{\max}$ is computed analytically by setting $\mu = 0$ and substituting $f_{t,ij}^{\prime \text{th}}$ into $h_{t,in'}^{A'}(f)$), finds the unique $\mu^\star, \nu^\star$ satisfying $R_t'(\mu^\star,\nu^\star) = f_{\max}^{S_t}$. Based on the above discussions, Algorithm 3 presents the solution of Problem $\mathcal{P}4$.

C. Complexity Analysis

We first analyze the complexity of the LDA scheme. The execution of LDA scheme consists of two parts: Offloading and resource optimization (lines 2-11 of Algorithm 1) and DNN parameters update (lines 12-20 of Algorithm 1). The offloading and resource optimization need to perform in every frame, while parameters update is performed infrequently. Therefore, we focus on analyzing the complexity of generating an offloading action in each frame. The major complexity is in line 8-10 of Algorithm 1, which executes Algorithm 2 to solve $\mathcal{P}3$ and $\mathcal{P}4$ A_t times in each frame.

Each BS solves the convex problem $\mathcal{P}3$ via a bisection search on the Lagrange multiplier λ. With $L_\lambda = \lceil \log_2(\lambda_{\max}/\varepsilon) \rceil$ iterations, for UEs of each BS solve (55) and (56) the per-iteration cost is $\mathcal{O}(J)$. Hence, the complexity of solving $\mathcal{P}3$ in Algorithm 2 is $\mathcal{O}\big(IJ\log(1/\varepsilon)\big)$ [52]. Problem $\mathcal{P}4$, adopts essentially the same bisection-based procedure as the BS, but operates a second bisection on an additional dual variable, this adds one extra logarithmic factor to the runtime, and its complexity is $\mathcal{O}\big(IJ\log^2(1/\varepsilon)\big)$. Because this term dominates $\mathcal{O}\big(IJ\log(1/\varepsilon)\big)$ cost already incurred for solving $\mathcal{P}3$ in Algorithm 2, the total cost of generating a single

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Algorithm 3: The computing resource optimization of LEOS for $\mathcal{P}4$

Input: $\{L_{t,ij}, T_{t,ij}^{S_t,proc}, Q_{ij}(t)\}_{i,j}, \phi, K, w, \kappa_2, \tau, f_{\max}^{S_t}, E_{\max}^{S_t}$

$\nu_{\mathrm{L}} \leftarrow 0,\ \nu_{\mathrm{H}} \leftarrow \nu_{\max}$;
while $|\nu_H - \nu_L| > \varepsilon$ **do**
  $\nu \leftarrow 0.5(\nu_{\mathrm{L}} + \nu_{\mathrm{H}})$;
  $\mu_{\mathrm{L}} \leftarrow 0,\ \mu_{\mathrm{H}} \leftarrow \mu_{\max}$;
  while $|\mu_H - \mu_L| > \varepsilon$ **do**
    $\mu \leftarrow 0.5(\mu_{\mathrm{L}} + \mu_{\mathrm{H}})$;
    compute $\{f_{t,ij}^{S_t}(\mu,\nu)\}$;
    compute $R_t'(\mu,\nu)$;
    if $R_t'(\mu,\nu) > f_{\max}^{S_t}$ **then** $\mu_{\mathrm{L}} \leftarrow \mu$;
    else $\mu_{\mathrm{H}} \leftarrow \mu$;
  end
  $E_t^{S_t} \leftarrow \kappa_2 \phi \sum_{i=1}^{I} \sum_j (f_{t,ij}^{S_t})^2 T_{t,ij}^{S_t,proc}$;
  if $E_t^{S_t} > E_{\max}^{S_t}$ **then** $\nu_{\mathrm{L}} \leftarrow \nu$;
  else $\nu_{\mathrm{H}} \leftarrow \nu$;
end
$\{f_{t,ij}^{S_t\,\star}\} \leftarrow \{f_{t,ij}^{S_t}(\mu_{\mathrm{H}}, \nu_{\mathrm{H}})\}$;
classify $\mathcal{A}'^{\star}, \mathcal{B}'^{\star}$ by $f_{t,ij}^{S_t\,\star}$ and $f_{t,ij}'^{\mathrm{th}}$;

Output: $\{\mathcal{A}'^{\star}, \mathcal{B}'^{\star}, f_{t,ij}^{S_t\,\star}\}$,

offloading decision in a frame is $\mathcal{O}\big(IJ\log^2(1/\varepsilon)\big)$. Therefore, the computational complexity of the LDA scheme in each frame is $\mathcal{O}\big(A_t IJ\log^2(1/\varepsilon)\big)$.

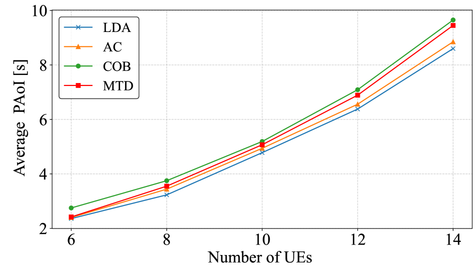
V. Simulation Results and Analysis

A. Parameters Setup

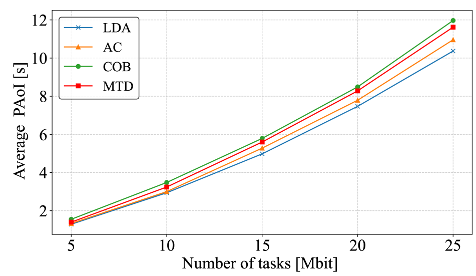
We consider an age-critical joint communication and computation offloading scenario that consists of LEOSs and three BSs. Each BS is 2 km apart, the heights of LEOSs are 200 km [5]. The operating frequency at BSs and the LEOS is 6 GHz and 30 GHz, respectively. The distances between UEs and BS are randomly distributed between 300 and 600 m [12], [38]. The number of tasks for UEs under each BS follows a normal distribution with a mean of L and a 3σ interval of $[L-5$ Mbit, $L+5$ Mbit$]$. The shadowed-Rician fading channel is with parameters $b_k = 0.158$, $\Omega_k = 1.29$, and $m_k = 20$ [46].

TABLE II
System Parameters

Parameter	Value	Parameter	Value
I	3	J	10
B_C	500 MHz	B_{Ka}	1 G Hz
$E_{\max}^{BS_i}$	120 J	$E_{\max}^{LEOS}$	60 J
$f_{BS_i}^{\max}$	4×10^9 CPU cycle/s	L	15 Mbit
$f_{LEO}^{\max}$	2×10^9 CPU cycle/s	ϕ	100
κ_1	2×10^{-26}	ΔT_A	100
κ_2	10^{-26}	τ	5 s
σ_1^2	2.2×10^{-12} w	γ	0.95
σ_2^2	1.37×10^{-18} w	ΔT_L	10
M	512	$\mathcal{C}$	32



(a) Average PAoI vs. number of UEs in each BS.



(b) Average PAoI vs. number of tasks.

Fig. 4. Average PAoI in comparison with LDA, AC, COB, and MTD schemes.

With reference to the setting of computing and communication parameters in system [12], [32], the main parameters in our system are set as in Table II.

In controlled experiments, we consider the following three schemes:

- Complete offloading to BS (COB) [12]: All the ground UEs offload their tasks to the BS for remote computing.
- Minimum transmission delay (MTD): Redirect the UE with the lowest uplink delay to the LEOS, while all others follow COB.
- Actor-Critic (AC) [37]: A constrained reinforcement learning agent that aims to minimize system PAoI in each frame, while explicitly incorporating both the accumulated backlog and energy consumption as constraints.

In contrast to the above three schemes, our LDA scheme is simulated, where the UEs offload their partial tasks either to the connected BS or LEOS.

B. Performance Evaluation

We evaluate the LDA scheme in terms of its system delay, and conduct a comparative analysis across three variables: local computational capability, the number of UE tasks, and the number of UEs. Unless otherwise stated, each metric is the mean over 10 independent training runs.

Fig. 4(a) shows PAoI rising with the number of UEs. It stays low and flat up to eight UEs, when all tasks still finish within one frame; after ten UEs backlog appears, and the penalty weight w accentuates the climb. Under heavier load, both the LDA and AC schemes offload more tasks to the LEOS, and their slope remains gentler than the COB and MTD schemes which rely on the BS. Fig. 4(b) echoes this pattern for total task volume: PAoI grows gradually while the BS is

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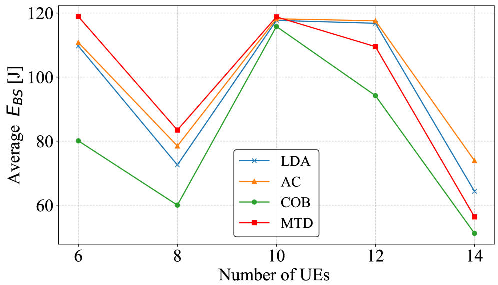
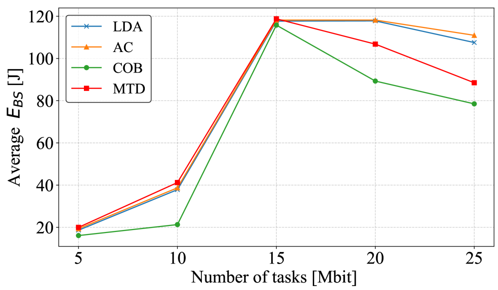
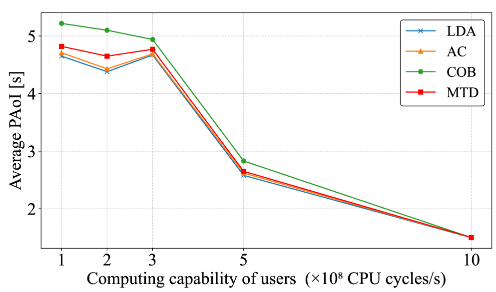


(a) BS energy consumption vs. number of UEs in each BS.

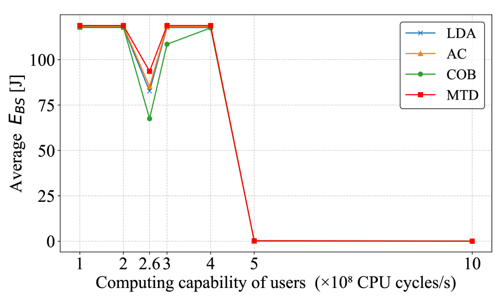


(b) BS energy consumption vs. number of tasks.

Fig. 5. Average BS energy consumption in comparison with LDA, AC, COB, and MTD schemes.



(a) Average PAoI vs. computational capability of UE.

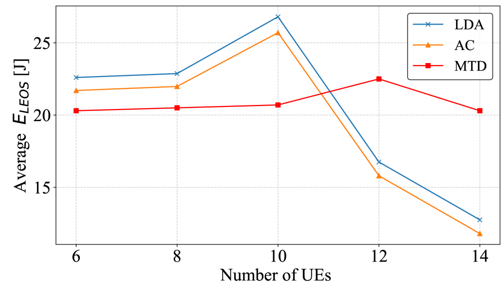


(b) BS energy consumption vs. computational capability of UE.

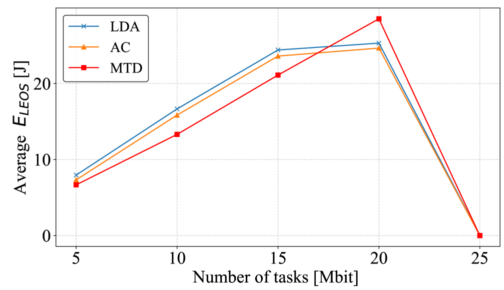
Fig. 6. System performance versus the computational capability of UE in comparison with LDA, AC, COB, and MTD schemes.

under-utilised, then sharply once saturation sets in. Dynamic offloading in the LDA and AC schemes can relief the BS burden, restraining PAoI relative to COB and MTD.

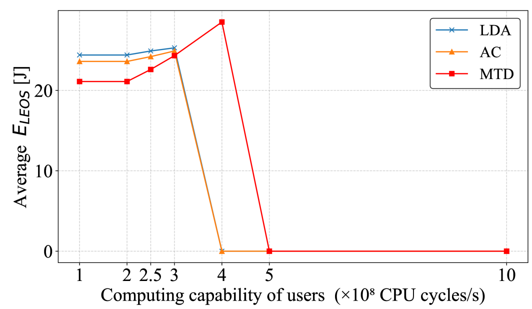
Fig. 5(a) plots total energy consumption versus the number of UEs. Because energy consumption grows quadratically with CPU frequency as shown in (20). Note that the COB scheme spreads a lower frequency across all tasks, leads to the decreasing of energy consumption up to eight UEs. Moreover,



(a) LEOS energy consumption vs. number of UEs in each BS.



(b) LEOS energy consumption vs. number of tasks.



(c) LEOS energy consumption vs. computational capability of UEs.

Fig. 7. Comparison between LDA, AC and MTD schemes, and the LEOS energy consumption versus different J and L and f_{ij}^{l}.

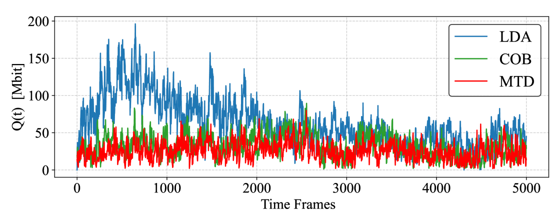


Fig. 8. The evolution of average data queue length over frames.

the MTD scheme leaves more work at the BS, needs higher frequencies, and therefore consumes more. A distinct peak at ten UEs arises across all schemes due to task overflow, which demands additional CPU resources in (50), leading to a sharp energy consumption surge. Beyond that point, the frequency of each UE falls again, and energy consumption drops. Fig. 5(b) mirrors this behaviour for growing task payloads. Energy consumption rises almost linearly below 10 Mbit, while tasks still finish in-frame; Once that threshold is crossed, overflows demand extra CPU.

Fig. 6(a) shows the effects of local computing threshold

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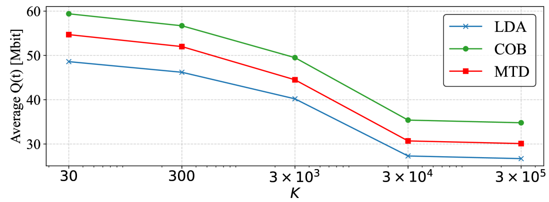


Fig. 9. Data queue vs. Lyapunov control parameter K.

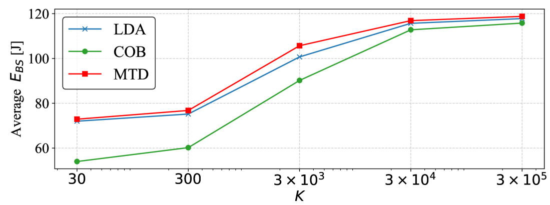


Fig. 10. BS energy consumption vs. Lyapunov control parameter K.

$T_{t,ij}^{l,proc}$ in (9). Below 3×10^8 cycles/s no task can finish locally, so everything is offloaded; latency then falls as the rising frequency lets more tasks stay on-device, reaching full local execution at 5×10^8 cycles/s. Fig. 6(b) relates BS energy consumption to the same frequency. Forced offloading at low frequency saturates the BS and maximises energy consumption. Around 2.5×10^8 cycles/s partial local processing cuts the load, lowering energy consumption; As local frequency approaches 3×10^8 cycles/s, more the tasks are processed locally, temporarily increasing per-task frequency at the BS and consequently BS energy consumption. After which complete local execution drives BS energy consumption to zero.

Fig. 7(a) plots LEOS energy consumption versus UEs. As OFDMA bandwidth per UE shrinks, transmission delay drives a pronounced energy consumption peak for MTD, after which the load shifts back to the BS and energy consumption falls. Both LDA and AC schemes begin slightly higher because they offload to the LEOS, but then decline steadily as the number of UEs grows. Fig. 7(b) varies data volume. The energy consumption of MTD scheme climbs almost linearly, offloading just one UE per BS regardless of size. In contrast, the LDA and AC schemes rise to the crest, and then taper off, showing that their adaptive offloading lightens the burden of LEOS at higher loads. Fig. 7(c) sweeps local CPU capability. Better CPUs keep small jobs on UE: The energy consumption of MTD scheme first rises and ultimately drops to zero; The LDA and AC schemes shed LEOS use sooner, hitting zero at a lower capability thanks to more aggressive local and BS processing.

Fig. 8 tracks the evolution of average data queue length of UEs over frames, excluding the AC scheme, which lacks queue constraints. We consider i.i.d. realizations of random events in 5,000 time frames, where each point in the figure is a moving-window average of 50 time frames. Initially, tasks accumulate due to random neural network decisions, increasing queue lengths. As the network trains and refines its offloading strategies, backlog is effectively cleared, reducing average

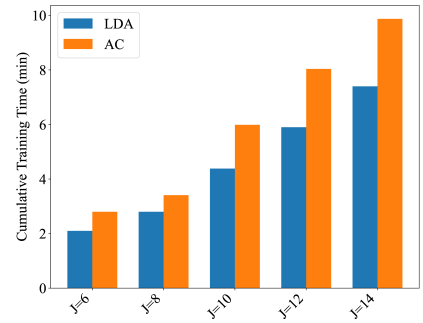
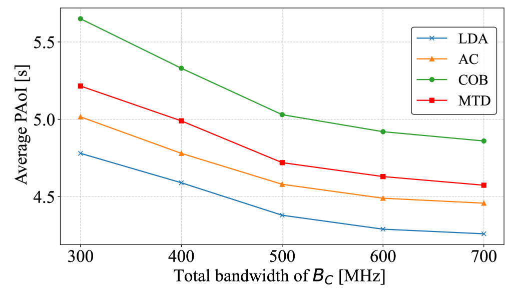
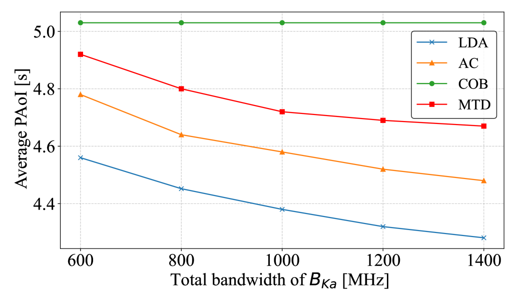


Fig. 11. Cumulative training time vs. number of UEs.



(a) Average PAoI vs. Total bandwidth of B_C.



(b) Average PAoI vs. Total bandwidth of B_{Ka}.

Fig. 12. Average PAoI versus the bandwidth of B_C and B_{Ka} in comparison with LDA, AC, COB, and MTD schemes.

queue length significantly. Upon reaching stable, optimized decisions, queue length stabilizes at a low and steady level, confirming effective long-term queue management.

Fig. 9 investigates the impact ofLyapunov control parameter K on data-queue backlog. We can observe that smaller K emphasizes energy consumption management, constraining immediate CPU allocations and causing queue growth. Increasing K shifts focus to timely task completion, reducing both PAoI and queue lengths. For large K, PAoI minimizes, and queues stabilize at low levels, clearly demonstrating the tradeoff management between energy consumption expenditure and queue stability.

Fig. 10 directly linked to Fig. 9, shows BS energy consumption against K. With smaller K, energy constraints dominate, restricting CPU frequencies and minimizing energy consumption. As K grows, the emphasis shifts towards task timeliness, reducing the weight of energy constraints, and thus increasing BS energy consumption. Eventually, with sufficiently large K, latency minimization prevails, driving energy consumption to

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the maximum permissible level, demonstrating a consistent, logical relationship with queue management illustrated in Fig. 9.

Fig. 11 contrasts the cumulative training time of the two pre-trained schemes over 100 rounds as the number of UEs per BS increases. The runtime of both methods grows almost linearly with J. However, the proposed LDA consistently outperforms the AC. Under less UEs ($J < 10$), LDA shortens the 100-round runtime by roughly 20%. As the number of UEs grows ($J \geq 10$), the saving enlarges to approximately 30%.

Fig. 12(a) and 12(b) show the impact of communication bandwidth on the average PAoI, analyzing the C-band for UE to BS uplinks and the Ka-band for UE to LEOS uplinks, respectively. Both figures demonstrate a monotonic decrease in PAoI as the available bandwidth increases. This trend is attributable to the higher achievable transmission rates, which directly reduce the uplink transmission delay. Notably, the performance improvement is most pronounced in the lower bandwidth regime. In this region, limited bandwidth creates a significant transmission bottleneck, leading to prolonged delays. Consequently, an increasing number of tasks cannot be completed within a single frame, resulting in substantial backlog and a sharp escalation in the weighted PAoI. Specifically, in Fig. 12(a), the schemes heavily reliant on the BS, such as COB and MTD, show the steepest decline. In contrast, Fig. 12(b) highlights the effectiveness of the LDA and AC schemes, which benefit significantly from increased LEOS bandwidth, whereas the performance of COB remains invariant as it does not utilize the satellite link.

VI. Conclusion

In this paper, we developed an age-critical joint communication and computation offloading scheme for satellite-integrated Internet, leveraging Lyapunov optimization to enhance long-term system performance. Specifically, we formulated a comprehensive optimization problem aimed at minimizing the average PAoI, subject to stringent energy consumption constraints on terrestrial BSs and LEOS. To effectively tackle this challenging long-term problem, we introduced a LDA scheme, which decomposes the optimization into manageable per-frame subproblems. Our LDA scheme employs a DNN integrated with an order-preserving quantization approach, significantly reducing computational complexity by efficiently decoupling offloading decisions from computational resource allocation. Simulation results validate that the proposed LDA scheme not only significantly improves the timeliness of task processing, but also ensures data queue stability and compliance with energy consumption constraints. These outcomes confirm the effectiveness and robustness of our LDA scheme in highly dynamic satellite-integrated Internet, highlighting its strong potential for future 6G applications. Future research will explore extending this framework to multi-satellite collaborative scenarios, addressing challenges such as inter satellite handovers and constellation level resource coordination.

Appendix A
Proof of (32), (33)

First, by squaring both sides of equation (26), we obtain:

$$\begin{aligned}Q_{ij}(t+1)^2 =& Q_{ij}(t)^2 + 2Q_{ij}(t)\left(L_{t,ij} - L_{t-1,ij}^{left} - L_{t,ij}^{proc}\right)\\ &+ \left(L_{t,ij} - L_{t-1,ij}^{left} - L_{t,ij}^{proc}\right)^2.\end{aligned} \tag{67}$$

Substituting it into (31) yields,

$$\begin{aligned}&0.5\sum_{i=1}^{I}\sum_{j=1}^{J} Q_{ij}(t+1)^2 - 0.5\sum_{i=1}^{I}\sum_{j=1}^{J} Q_i(t)^2\\ &= 0.5\sum_{i=1}^{I}\sum_{j=1}^{J}\left(L_{t,ij} - L_{t-1,ij}^{left} - L_{t,ij}^{proc}\right)^2\\ &+ \sum_{i=1}^{I}\sum_{j=1}^{J} Q_{ij}(t)\left(L_{t,ij} - L_{t-1,ij}^{left} - L_{t,ij}^{proc}\right).\end{aligned} \tag{68}$$

For the squared term in this formula, replace $L_{t-1,ij}^{left}$ with $(L_{t-1,ij} - L_{t-1,ij}^{proc})$. Given the (13), we can get:

$$(L_{t,ij} + L_{t-1,ij}^{proc} - L_{t-1,ij} - L_{t,ij}^{proc})^2 \leq (max\{L_{t,ij}, L_{t-1,ij}\})^2. \tag{69}$$

Thus, the proof of (32) and (33) are completed.

Appendix B
Proof of (36), (37)

Similar to the Appendix A, by squaring both sides of equation (27), we obtain:

$$\begin{aligned}E_i(t+1)^2 =& E_i(t)^2 + 2E_i(t)\left(E_t^{BS} - E_{max}^{BS}\right)\\ &+ \left(E_t^{BS} - E_{max}^{BS}\right)^2.\end{aligned} \tag{70}$$

Substituting it into (35) yields,

$$\begin{aligned}&0.5\sum_{i=1}^{I} E_i(t+1)^2 - 0.5\sum_{i=1}^{I} E_i(t)^2\\ &= 0.5\sum_{i=1}^{I}\left(E_t^{BS_i} - E_{max}^{BS_i}\right)^2 + \sum_{i=1}^{I} E_i(t)\left(E_t^{BS_i} - E_{max}^{BS_i}\right).\end{aligned} \tag{71}$$

By the CPU power model at the BS, the per-frame energy satisfies $E_t^{BS_i} \leq \kappa_1 \left(f_{\max}^{BS_i}\right)^3 \tau$, when the maximum CPU frequency is used for the whole frame of length τ. Hence, for every i and t,

$$\left(E_t^{BS_i} - E_{\max}^{BS_i}\right)^2 \leq \max\Big\{\left(\kappa_1 (f_{\max}^{BS_i})^3 \tau\right)^2,\ \left(E_{\max}^{BS_i}\right)^2\Big\}. \tag{72}$$

For the squared term in this formula, substituting the maximum value of $E_t^{BS_i}$ into the expression, we can get:

$$C_2 \triangleq \tfrac{1}{2}\sum_{i=1}^{I} \max\Big\{\left(\kappa_1 (f_{\max}^{BS_i})^3 \tau\right)^2,\ \left(E_{\max}^{BS_i}\right)^2\Big\}.$$

Substituting the bound into the drift expression gives

$$\Delta V(\mathbf{E}(t)) \leq C_2 + \sum_{i=1}^{I} E_i(t)\,\mathbb{E}\Big\{E_t^{BS_i} - E_{\max}^{BS_i}\,\Big|\,\mathbf{X}(t)\Big\}.$$

Thus, the proof of (36) and (37) are completed.

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